Unit 3 · Outcome 2 · Sustainable Development

The beneficial and harmful impacts of a case study on Earth's four interrelated systems

KK07 VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. Earth systems thinking Beneficial & harmful 5 practice questions · 20 marks ◆ The exact dot point

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.

The beneficial and harmful impacts of a case study on Earth's four interrelated systems

U3·O2 · KK07 Sustainable Development The beneficial and harmful impacts of a case study on Earth's four interrelated systems Pull one thread of an ecosystem and the whole web moves. This dot point asks you to take your case study — here, the Boorla Wetlands recovery — and trace how its impacts spread, for better and for worse, through the atmosphere , biosphere , hydrosphere and lithosphere . Examinable key knowledge Earth systems thinking Beneficial & harmful 5 practice questions · 20 marks ◆ The exact dot point "The beneficial and harmful impacts of th

01 The four systems, defined

01 The four systems, defined Environmental scientists model Earth as four interrelated systems . Every environmental impact lands in one or more of them. Know the one-line definition of each cold — examiners give easy marks for it, and you can't trace an impact if you can't name the system it lands in. Atmosphere AIR & GASES The layer of gases surrounding Earth — and everything carried in it: oxygen, carbon dioxide, methane, water vapour , dust and heat. Think weather, greenhouse gases, air quality and microclimate. Biosphere LIVING THINGS All the region

02 The big idea: systems are interrelated

02 The big idea: systems are interrelated This is the heart of the dot point — and the place most marks are won or lost. The four systems are not four separate boxes. Matter and energy move between them constantly , so a change in one cascades into the others. VCAA calls this Earth systems thinking , and the command verbs ("analyse", "evaluate") demand you show the links , not just list impacts in silos. ◆ What "interrelated" buys you in the exam A weak answer says: "Watering the wetland helps the plants." One system, no link, ~1 mark. A strong answer tr

03 The impacts ledger — beneficial & harmful

03 The impacts ledger — beneficial & harmful For the exam you need a stocked toolkit: at least one beneficial and one harmful impact for each system, with the mechanism (the "because…"). Here is the Boorla ledger. Notice the recurring theme — the same restoration is a gift to some systems and a cost to others, and even within one system it can be good long-term but bad short-term. Hydrosphere WATER river · wetland · groundwater · salinity + Wetland re-fills and the natural wetting–drying cycle returns; the water table recharges. + Diluted salinity and

04 No system stands alone

04 No system stands alone Tap any system in the web below. Watch its links to the other three light up, and the floating + / – tags show how a single change pushes and pulls everything connected to it. This is the visual proof that you can't quarantine an impact to one box. Interrelation web · tap a system tap a node to trace its links Hydrosphere Biosphere 🌤 Atmosphere ▲ Lithosphere Tap a system to see what it touches.

05 Practice: sort the impact

★ PRACTICE & SELF-MARK

05 Practice: sort the impact The single most-tested micro-skill here: take an impact statement and (1) name the system it lands in, then (2) call it beneficial or harmful . Work through all eight. Instant feedback; your score feeds the dock. — Step 1 — which system? Step 2 — beneficial or harmful? 0 / 8 Next →

06 Decode the command word

◈ COMMAND WORDS

06 Decode the command word The verb tells you how hard to work. "Identify the system" wants one word; "evaluate the impacts" wants a weighed judgement across systems. Tap a verb to see what it demands for this dot point . Identify / State Describe Explain Analyse Evaluate Distinguish

07 Build an answer: the P·E·L trainer

✎ P·E·L WRITING

07 Build an answer: the P·E·L trainer For an "impacts on Earth's systems" answer, the reliable structure is P·E·L : Point (name the system + beneficial/harmful), Example (the specific Boorla mechanism), Link (cascade it to another system). Draft each part, then reveal the model. Prompt: Describe one harmful impact of the Boorla environmental flow on the atmosphere, and link it to another system. (3 marks) P — POINT Name the system & the direction (harmful). E — EXAMPLE The specific Boorla mechanism. L — LINK Cascade it to another system. Reveal model ans

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

08 Six exam traps

⚠ EXAM TRAPS

08 Six exam traps T1 Listing impacts without naming the system "It helps the plants and the water." Which sphere? Always pin each impact to atmosphere / biosphere / hydrosphere / lithosphere by name. T2 Only beneficial, or only harmful If the question says "beneficial and harmful", a one-sided answer caps your marks. Give both, even when one feels obvious. T3 Treating systems as silos — the big one The dot point says interrelated . Higher marks require you to link at least two systems (the cascade), not list four disconnected facts. T4 Water vapour in th

09 One-screen cheat sheet

✦ RECAP / CHEAT SHEET

09 One-screen cheat sheet The four systems & your go-to Boorla impacts Hydrosphere — water + wetland refills, salinity diluted, water table recharged · – less water for town/farms, blackwater drops O_2 Biosphere — life + red gums, waterbirds, native fish & microbes return · – fish kills (hypoxia), carp & mozzies benefit 🌤 Atmosphere — air + carbon sink, cooler/humid microclimate, less dust · – methane from anaerobic sediment, CO_2 pulse ▲ Lithosphere — land + less erosion, soil rebuilds, dryland salinity eased · – acid sulfate soils, bank erosion, salt

10 Exam practice · weak vs strong

✗ WEAK ANSWER

weak response, the annotated

✓ STRONG ANSWER & MARKING

strong response and the marking guide. Mark your own attempt — the dock keeps your running total.

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.